Indian Institute of Technology Jodhpur
Office of Academics
NH65, Nagaur Road, Karwar, Jodhpur 342037, INDIA

The course and timings

Course Number and Title	MAL1020 Mathematics II				
Lectures	**English**	Days	Monday	Tuesday	Thursday
		Time	1:00 PM	1:00 PM	1:00 PM
		Venue	LHC 308 and LHC 110		
	Non-English	Days	Monday	Tuesday	Thursday
		Time	4 PM	8 AM	8 AM
		Venue	LHC 204 and 205		
Tutorials		Days	To be announced		
		Time	–		
		Venue	–		
Laboratory		Days	NA		
		Time	NA		
		Venue	NA		

Instructors

Name	Email	Department
Dr. Abhishek Sarkar	abhisheks@iitj.ac.in	Mathematics
Dr. Pradip Sasmal	psasmal@iitj.ac.in	Mathematics

Office timings during the week: Office hours: Tuesday, 4-5 pm, Department of Mathematics

Contact details of Teaching Assistants (TAs): For any queries related to the course or attendance, please contact the TAs directly.

Name of the TAs	Contact Details
Nishtha Tomar	P21MA204
Subhadip Sahoo	P25MA0013
Barsha Shaw	P22MA202
Ranit Dutta	P22MA207
Sanu Sarkar	P24MA0204
Shubham Kumar Jain	P22MA208
Arijit Saha	P23MA0005
Geetanjli	P23MA0007
Harshit Jain	P23MA0008

Jaikishan Bairwa	P25MA0006
Vishnu Kumar	P25MA0014
Adarsh Vishwakarma	P25MA0001
Akash Samanta	P25MA0002
Krishnendu Das	P25MA0007
Preetam Joshi	P25MA0009
Prateek Somvanshi	P25MA0008
Indranil Mukherjee	P25MA0005
Shivam Kumar	P24MA0006
Anshul Jain	P25MA0003
Somnath Kumar	P25MA0012
Swaroop Dan Charan	P24MA0002
Radhika Sharma	P24MA0203

Lecture-wise detailed course content as approved by the Senate:

Title	**Mathematics II**	**Number**	**MAL1020**
Department	**Mathematics**	**L-T-P [C]**	**3-1-0 [4]**
Offered for	**B.Tech. 1st Year**	**Type**	**Compulsory**
Prerequisite	**None**		

Objectives
The Instructor will:
1. Provide an understanding of the fundamentals of Linear Algebra.
2. Provide understanding of the fundamentals of Differential Equations.

Learning Outcomes
The students are expected to have the ability to
1. Impart knowledge of different structures and their properties, like dependence, basis, and dimension.
2. Gain a strong understanding of linear transformations between structures and their relation with matrices.
3. Solving analytical and series solutions for ordinary differential equations and analyzing the stability of linear ODE systems.

Contents

Linear Algebra (20 Lectures):
[10 Lectures] Matrices, Vector spaces (over the field of real and complex numbers), subspaces, linear dependence/ independence, basis, dimensions, coordinate with respect to a basis, complementary subspaces, Linear transformations, Range space and rank, Null space and nullity, matrix representation of linear transformation, change of basis and similarity, rank-nullity theorem.
[10 Lectures] Inner product, Norm, Gram-Schmidt orthogonlization process, orthonormal bases. Eigenvalues and eigenvectors, characteristic polynomials, Cayley-Hamilton theorem, properties of eigenvalues and eigenvectors, diagonalization of matrices.

Differential Equations (20 Lectures):
[8 Lectures] First Order Ordinary Differential Equations: Geometrical interpretation of solution, Solution methods for separable equations, Exact equations, Linear equations, Existence and uniqueness of solution of IVP, Wronskian and general solution of nonhomogeneous equations, Picard's Theorem for IVP (without proof). Second Order Linear differential equations: General solution of homogeneous equation, Euler-Cauchy Equation, Extensions of the results to higher order linear differential equations.
[12 Lectures] Power Series Method, Frobenius Method, S.L.P. and orthogonal functions, System of first order linear ODE and its stability.

Textbook

1. Axler, S. (1997). Linear Algebra Done Right (Undergraduate Texts in Mathematics), Springer.
2. Kreyszig, E. (2010). Advanced Engineering Mathematics, John Wiley & Sons.
3. Simmons, G.F. (2002). Differential Equations with Applications and Historical Notes, Tata McGraw-Hill.
4. Boyce, W.E. and DiPrima, R.C. (2006). Elementary Differential Equations and Boundary Value Problems, John Wiley & Sons.

Reference Books

1. Strang, G. (2006). Linear Algebra and Its Applications, Thomson, Brooks/Cole.
2. Kumaresan, S. (2000). Linear Algebra - A Geometric Approach, PHI Learning.
3. Coddington, E.A. (1961). An introduction to ordinary differential equations, Prentice-Hall.
4. Lipschutz, S. and Lipson, M. (2012). Schaum's Outline of Linear Algebra, McGraw-Hill.
5. Bronson, R. and Costa, G. (2010). Schaum's Outline of Differential Equations, McGraw-Hill.

Online Course Material

1. Rana, I.K. (IIT Bombay), Basic Linear Algebra, https://nptel.ac.in/courses/111101115/
Manam, S. (IIT Madras), Differential Equations for Engineers, https://nptel.ac.in/courses/111106100/1

Evaluation policy for the course:

Components	Weightage (%)	Date and Timings	Remarks
Assignments	10%	It will be mentioned separately on Google Classroom.	The number of assignments is 4.

Quizzes	20%	The announcement will be made in class and posted on Google Classroom.	No make-up will be arranged. The best two will be considered out of three.
Minor	25%	As per the Academic Calendar	Written Examination
Major	45%	As per the Academic Calendar	Closed Books, written examinations, and scientific calculators will not be permitted

Guidelines for Examination: As per the Institute's rules and regulations.

Attendance Policy: As per institute norms, a student is expected to have full attendance in each course unless the student takes a leave of absence for valid medical or bona fide reasons. In any case, the attendance of a student should not fall below 75% in a course. The attendance records must be made available weekly to all students enrolled in a course by the instructors/TAs.

1. If a student remains absent from a class without sanctioned leave for more than two weeks, then the course instructor may recommend the de-registration of the student from the course.
2. If a student is found to be absent from all academic activities for four or more weeks (not necessarily contiguous) in a semester, with or without sanctioned leave, then his/her registration will result in automatic semester withdrawal for all the courses.
3. If a student's attendance falls below 75% without any valid reason, the instructors can recommend de-registration from the course to the SUGC/SPGC.

Any other relevant information: For any queries, you can visit our office during the mentioned office hours. Moreover, you can also post your doubts and queries on Google Classrooms.